

loss f^n : regularized
differentiable.

XGBoost for Regression

cgpa	Package
6.7	4.5
9.0	11.0
7.5	6.0
5.0	8.0

Working principle
same as Gradient
Boosting.

But, in Xg boost the decision tree is different.

cgpa	package	pred model 1	residual 1	pred model 2	residual 2
6.7	4.5	7.3	-2.8	6.69	-2.19
9.0	11.0	7.3	3.7	8.41	2.59
7.5	6.0	7.3	-1.3	6.69	-0.69
5.0	8.0	7.3	0.7	7.51	0.19

residual
is decreasing

mean

decision tree by cgpa + Residual 1.

$[-2.8, 3.7, -1.3, 0.7]$ → Similarity score (SS)

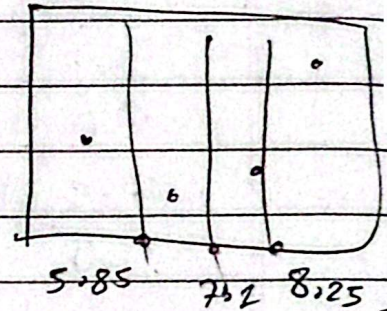
$$SS = \frac{(\sum \text{residuals})^2}{\# \text{residuals} + \lambda} \rightarrow \text{regularization.}$$

$$L=0$$

$$SS_{\text{root}} = \frac{(-2.8 + 3.7 - 1.3 + 0.7)^2}{4+0} = \frac{0.3^2}{4} = 0.02$$

Now, sort the feature in ascending.

$\left. \begin{array}{l} 5.0 \\ 6.7 \\ 7.5 \\ 9.0 \end{array} \right\} \begin{array}{l} 5.85 \\ 7.1 \\ 8.25 \end{array} \left. \vphantom{\begin{array}{l} 5.0 \\ 6.7 \\ 7.5 \\ 9.0 \end{array}} \right\} \text{splitting criteria.}$
 means of 2



which gives better ~~output~~ ~~result~~ improvement becomes root node.

Let $L=0$ for new.

Splitting criteria

① 5.85

$$c_{\text{gpa}} < 5.85$$

$$\text{Left } ① \quad 0.7$$

$$\text{Right } ② \quad -2.8, -1.3, 3.7$$

$$SS = \frac{(\sum \text{residuals})^2}{\# \text{residual} + 1}$$

$$SS_L = \frac{0.7^2}{1+0} = 0.49$$

$$SS_R = \frac{(-2 - 1.3 + 3.7)^2}{3+0} = 0.55$$

$$\text{gain} = (SS_L + SS_R) - SS_{\text{root}} = 0.52$$

Splitting Criteria

② 7.1

$$c_{\text{gpa}} < 7.1$$

$$\text{Left } ① \quad 0.7, -2.8$$

$$\text{Right } ② \quad -1.3, 3.7$$

$$SS_L = \frac{(0.7 - 2.8)^2}{2+0} = 2.20$$

$$SS_R = \frac{(-1.3 + 3.7)^2}{2} = 2.88$$

$$\text{gain} = (SS_L + SS_R) - SS_{\text{root}} = 5.06$$

Splitting Criteria

③ 8.25

$$c_{\text{gpa}} < 8.25$$

$$\text{Left } ① \quad 6.7, -2.8, -1.3$$

$$\text{Right } ② \quad 3.7$$

$$SS_L = 3.85$$

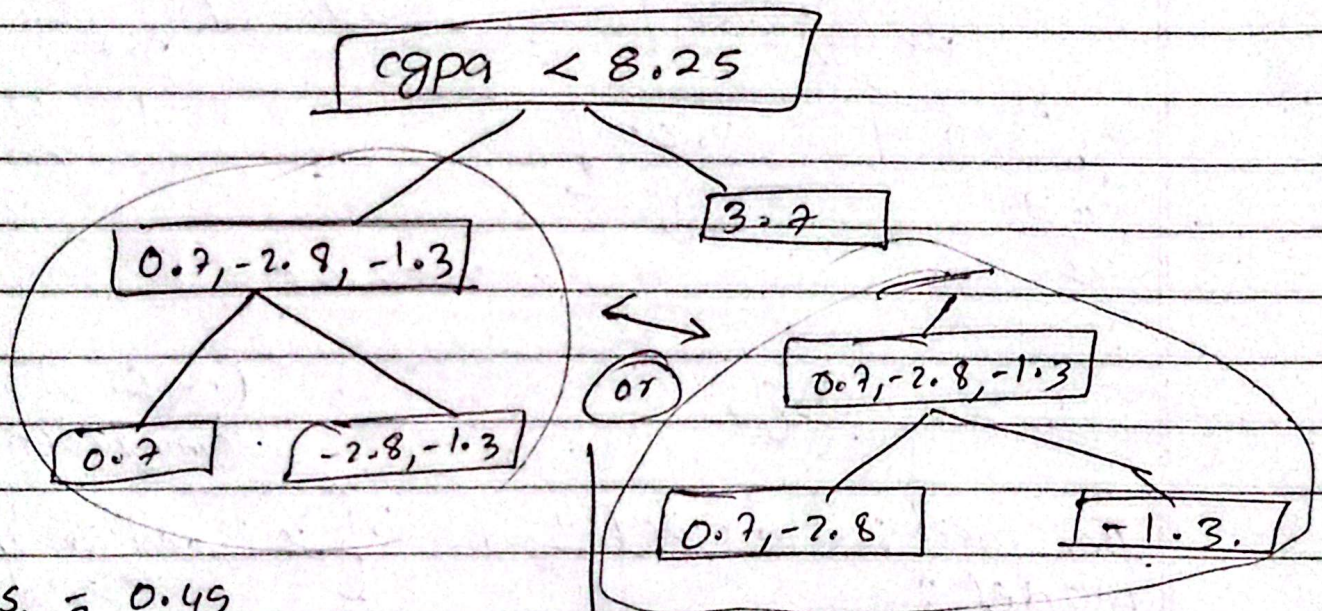
$$SS_R = 13.65$$

$$\text{gain} = 17.52$$

for stage two, root node will be.

Criteria (3) $cgpa < 8.25$

cos it has higher gain



$$SS_L = 0.49$$

$$SS_{root} = 3.85$$

$$SS_R = 8.40$$

$$gain = 5.04$$

$$SS_L = 2.20$$

$$SS_R = 1.09$$

$$SS_{root} = 3.85$$

$$gain = 0.04$$

Choose this one.

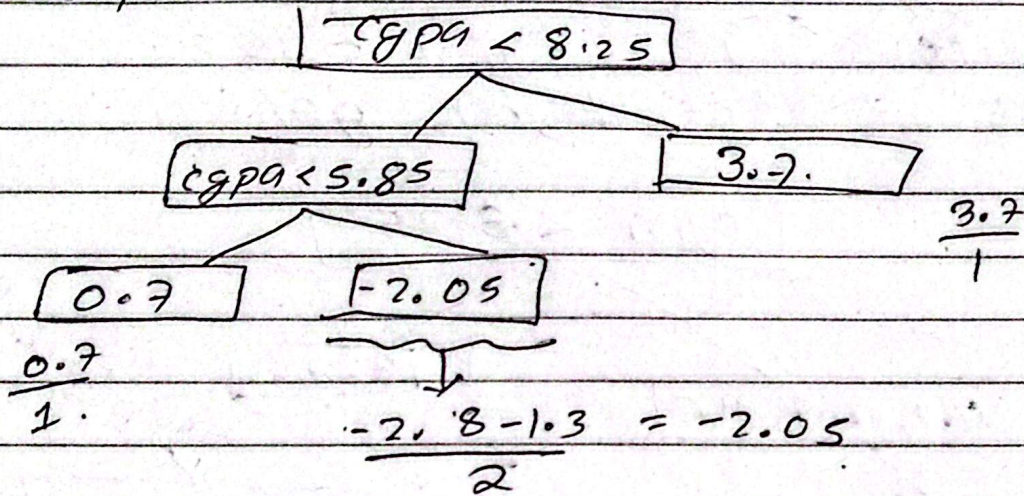


$$\text{Output} = \frac{\text{sum of residual}}{\# \text{ residual} + L}$$

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For output:



$L=0$

for now:

avg:

(~ 0.3)

learning rate

For pred: model 1 + η x decision tree's output.

for model 3: model 2 + η x new output.